12. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines.

(1) 6 (2) 9 (3) 12 (4) 18

13. Everybody in a room shakes hands with everybody else. The total number of shake hands is 66. The number of persons in the room is $\cdots\cdots\cdots\cdots$

(1) 11 (2) 12 (3) 10 (4) 6

14. Number of sides of a polygon having 44 diagonals is $\cdots\cdots$

(1) 4 (2) $4!$ (3) 11 (4) 22

15. If 10 lines are drawn in a plane such that no two of them are parallel and no three are concurrent, then the total number of points of intersection are

(1) 45 (2) 40 (3)10! (4) 2^{10}

16. In a plane there are 10 points are there out of which 4 points are collinear, then the number of triangles formed is

(1) 110 (2) ${}^{10}C_3$ (3) 120 (4) 116

17. In ${}^{2n}C_3 :^n C_3 = 11 : 1$ then n is

(1) 5 (2) 6 (3)11 (4)7

18. ${}^{(n-1)}C_r +^{(n-1)} C_{(r-1)}$ is

(1) ${}^{(n+1)}C_r$ (2) ${}^{(n-1)}C_r$ (3) ${}^nC_r$ (4)${}^nC_{r-1}$.

19. The number of ways of choosing 5 cards out of a deck of 52 cards which include at least one king is

(1) ${}^{52}C_5$ (2) ${}^{48}C_5$ (3)${}^{52}C_5 +^{48} C_5$ (4)${}^{52}C_5 -^{48} C_5$.

20. The number of rectangles that a chessboard has $\cdots$

(1) 81 (2) 9^9 (3)1296 (4) 6561

21. The number of 10 digit number that can be written by using the digits 2 and 3 is

(1) ${}^{10}C_2 +^9 C_2$ (2) 2^{10} (3)$2^{10} - 2$ (4) 10!

22. If P_r stands for ${}^rP_r$ then the sum of the series $1 + P_1 + 2P_2 + 3P_3 + \cdots + nP_n$ is

(1) P_{n+1} (2) $P_{n+1} - 1$ (3) $P_{n-1} + 1$ (4)${}^{(n+1)}P_{(n-1)}$

23. The product of first n odd natural numbers equals

(1) ${}^{2n}C_n \times^n P_n$ (2) $(\frac{1}{2})^n \times {}^{2n}C_n \times^n P_n$ (3)$(\frac{1}{4})^n \times {}^{2n}C_n \times^{2n} P_n$ (4)${}^nC_n \times^n P_n$

24. If ${}^nC_4$, ${}^nC_5$, ${}^nC_6$ are in AP the value of n can be

(1) 14 (2) 11 (3)9 (4)5

25. $1 + 3 + 5 + 7 + \cdots + 17$ is equal to

(1) 101 (2) 81 (3) 71 (4) 61